```
from google.colab import files

uploaded = files.upload()
```

Add text cell

Choose files half wave dipole.png

half wave dipole.png(image/png) - 164266 bytes, last modified: 12/05/2025 - 100% done

```
Saving half wave dipole.png to half wave dipole.png
```

```
import cv2
import matplotlib.pyplot as plt

# Read the uploaded image
image_name = list(uploaded.keys())[0]
img = cv2.imread(image_name)

# Convert the image to grayscale
gray_img = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Apply histogram equalization
equalized_img = cv2.equalizeHist(gray_img)

# Display original grayscale image
plt.subplot(1, 2, 1)
plt.imshow(gray_img, cmap="gray")
plt.title("Original Image")
plt.axis("off")

# Display equalized image
plt.subplot(1, 2, 2)
plt.imshow(equalized_img, cmap="gray")
plt.title("Histogram Equalized Image")
plt.axis("off")

plt.show()

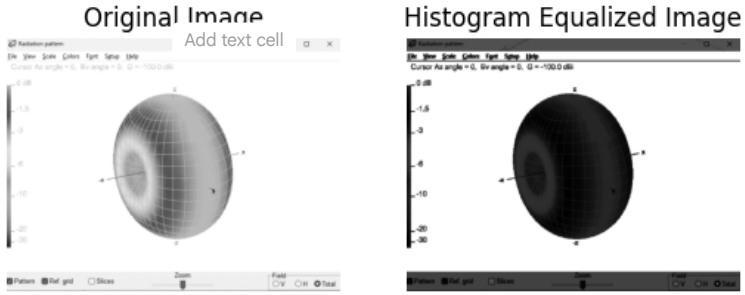
# Display histograms for comparison
plt.figure(figsize=(10, 4))

plt.subplot(1, 2, 1)
plt.hist(gray_img.ravel(), 256, [0, 256])
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

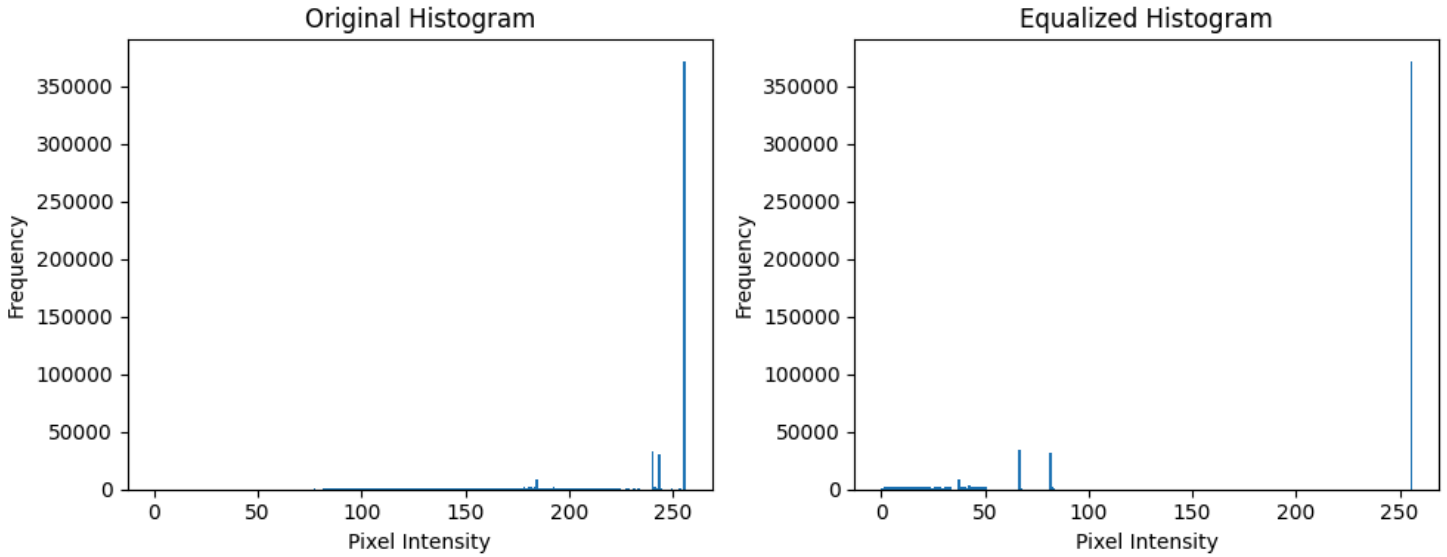
plt.subplot(1, 2, 2)
plt.hist(equalized_img.ravel(), 256, [0, 256])
plt.title("Equalized Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
```

```
plt.ylabel('Frequency')

plt.tight_layout()
plt.show()
```




```
/tmp/ipykernel_2344/458196584.py:32: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is deprecated since Matplotlib 3.9; the parameter
  plt.hist(gray_img.ravel(), 256, [0, 256])
/tmp/ipykernel_2344/458196584.py:38: MatplotlibDeprecationWarning: Passing the range parameter of hist() positionally is deprecated since Matplotlib 3.9; the parameter
  plt.hist(equalized_img.ravel(), 256, [0, 256])
```



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